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An integrated analysis of Eco–Geo–Hydro processes in a wetland on Jeju Island, South Korea

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ABSTRACT
Wetland terrestrialization has been increasingly recognized as a process shaped by coupled interactions among hydrological change, peat accumulation, and hydrarch succession. However, conventional water-budget approaches for predicting terrestrialization in individual wetlands have focused on external hydrological inputs and losses, while internal feedbacks associated with peat accumulation and hydrarch succession have not been explicitly incorporated. This study developed an Eco–Geo–Hydro water-budget model for Seoyeongari wetland, a mountain wetland on Jeju Island, Korea, and evaluated its empirical basis using field investigations. Airborne image analysis, vegetation surveys, dendrochronological analysis, and soil organic matter assessment showed the co-occurrence of pond area reduction, a vegetation gradient from aquatic plants to terrestrial vegetation, tree encroachment, and spatial differences in soil organic matter content. Based on these observations, four model scenarios were constructed according to whether peat accumulation and hydrarch succession were incorporated, and simulated pond area changes from 2011 to 2023 were compared with observations. The Integrated model, which incorporated both peat accumulation and hydrarch succession, most closely reproduced the observed pond area changes. This suggests that peat accumulation and hydrarch succession can influence terrestrialization by altering storage capacity and evapotranspiration. Therefore, predicting long-term changes in individual wetlands requires consideration of not only external hydrological conditions but also internal ecogeomorphological and ecohydrological processes. This study demonstrates that terrestrialization should be understood not simply as pond area reduction, but as a process in which ecological, geomorphological, and hydrological processes interact to determine the future state of wetlands.

Keywords: Water budget model, Wetland, Terrestrialization, Hydrarch succession, Peat accumulation

1. Introduction

Until the mid-twentieth century, terrestrialization was generally interpreted as an outcome of hydrarch succession (Cowles, 1901; Gates, 1912; Clements, 1916). In this classical framework, Clements' (1916) concept of hydrosere, combined with the idea of autogenic succession, served as a field-based explanatory framework for wetland development involving the accumulation of organic sediments (Pearsall, 1920; Wilson, 1935). However, as it became increasingly recognized that the rates and pathways of succession are profoundly shaped by environmental forces—such as hydrology, climate, and disturbance—as well as by resource limitations (Drury and Nisbet, 1973; van der Valk, 1981), more comprehensive frameworks, such as the integrated model of raised-wetland development, began to emerge. This integrated approach combines theories of wetland hydrology (Ingram, 1982), peat growth (Clymo, 1978, 1984), and the differentiation of surface features resulting from hydrology and spatial patterns of peat accumulation (Foster and Wright, 1990). At the site scale, water-budget models provide a practical hydrological approach for simulating wetland dynamics and have been successfully applied to many individual wetlands, including those in temperate regions, at much finer spatial resolutions (Mokhesuoe et al., 2023; Oh et al., 2018; Wallace et al., 2024; Yang et al., 2009).

Nevertheless, their predictive scope often excludes key ecohydrological and ecogeomorphological feedbacks, particularly vegetation dynamics and peat-forming processes. Previous studies have noted that conventional water-budget approaches often fail to account for vegetation-driven evapotranspiration and internal wetland dynamics, particularly peat accumulation and decomposition (Holden et al., 2004; Mokhesuoe et al., 2023; Seo et al., 2025). Belyea and Baird (2006) further argued that peatland development models may lead to misleading interpretations if they fail to incorporate interactions between ecological and hydrological processes.

Despite these calls for integration, few water budget analyses have incorporated such ecohydrological and ecogeomorphological feedbacks to predict wetland terrestrialization. Ecogeomorphological feedbacks proposed for peatlands include a negative feedback in which peat accumulation causes peripheral areas to be rapidly raised above the water table, allowing woody plants to encroach and increasing the rate of organic carbon decomposition (Drollinger et al., 2020; Zhang et al., 2025), and a positive feedback in which peat accumulation leads to a rise in the water table and consequently reduces the rate of organic carbon decomposition (Ise et al., 2008; Regan et al., 2019). Ecohydrological feedbacks in peatlands operate as follows: when a decline in water level occurs, 나지가 드러나고, 식생이 침입하면서 woody plants expand, and the increase in evapotranspirative water loss caused by woody vegetation reduces water availability(Pauliukonis and Schneider, 2001). Such potential positive feedbacks of woody vegetation to autogenic drying may accelerate tree encroachment (Budny and Benscoter, 2016; Moore et al., 2022; Zhong et al., 2020). Although these two feedback mechanisms have been extensively investigated individually through theoretical, experimental, and modeling studies, integrated studies that apply both mechanisms simultaneously to individual wetland systems remain limited.

Therefore, this study develops a parsimonious Eco–Geo–Hydro model to integrate hydrological change, peat accumulation, and hydrarch succession, all of which influence terrestrialization in individual wetlands. Whereas conventional water-budget models primarily simulate wetland water-level and area dynamics as a function of external hydrological inputs and losses, the proposed model explicitly incorporates two internal feedback mechanisms that can influence terrestrialization: peat accumulation, which alters storage capacity, and hydrarch succession, which alters evapotranspiration. However, before applying this model, it is necessary to establish empirical evidence for ongoing terrestrialization in the target wetland and to assess whether vegetation distribution, peat accumulation, and hydrological conditions are interrelated. Accordingly, this study applied the proposed framework to Seoyeongari wetland, a mountain wetland on Jeju Island, South Korea, using airborne image analysis, vegetation surveys, dendrochronological analysis, and soil organic matter assessment to evaluate the progression of terrestrialization and the interrelationships among ecological, geomorphological, and hydrological processes. Finally, by comparing the performance of four model scenarios that differ in whether peat accumulation and hydrarch succession are incorporated, this study evaluates whether coupling these processes with a water-budget model contributes to explaining wetland area reduction and terrestrialization in an individual wetland.

2. Materials and methods

2.1. Study area

The study area, Seoyeongari wetland, is located on Jeju Island, South Korea (33°19′36.9″N, 126°23′50.3″E), at an elevation of 634 m above sea level (Figs. 1a, b). Seoyeongari wetland is situated within a crater on the southwestern slope of a small monogenetic volcanic cone. The surrounding slope has a gradient of approximately 15–30%, and the mapped soil unit is the Jungmun series. The Jungmun series is classified as an Acrudoxic Fulvudand, belonging to the Andisols, and is derived from pyroclastic materials. It is generally characterized by well-drained conditions, moderate permeability, medium to rapid runoff, and deep soil development, with an effective soil depth exceeding 100 cm. Although these volcanic-ash-derived soils are generally well drained, wetland formation in crater settings on Jeju Island can occur where fine materials accumulate on the crater floor through slope degradation and/or where local subsurface impervious layers restrict drainage, as reported for Mulyeongari wetland (Kim, 2009). Therefore, the occurrence of Seoyeongari wetland within an otherwise well-drained volcanic setting may be explained by localized geomorphic and subsurface conditions that reduce drainage at the crater floor.

Meteorological records from the nearest Seogwang-ri automatic weather station (33°18′16.4″N, 126°18′21.6″E) showed that the mean annual air temperature and precipitation from 2011 to 2023 were 15.3 °C and 1725.4 mm yr^{-1} (Data sources are listed in Supplementary Table A.1). The area surrounding Seoyeongari wetland is covered by 30–40-year-old *Cryptomeria japonica* plantations and other broad-leaved forests. According to historical records, the study area was originally grassland until the mid-twentieth century; however, land cover changed to forest following afforestation projects implemented in the 1970s (Kim et al., 2025).

2.2. Field surveys

The field surveys assessed~~were conducted to assess~~ whether terrestrialization is occurring in Seoyeongari wetland through an integrated approach combining airborne image analysis, vegetation surveys with dendrochronological analysis, and soil analysis (Supplementary Table A.1). Together, these surveys establish whether ecological, geomorphological, and hydrological processes need to be considered jointly.~~Furthermore, these surveys provided the basis for evaluating the need for an integrated analysis of ecological, geomorphological, and hydrological processes.~~

To assess whether the area of open-water ponds has decreased as a result of terrestrialization, airborne image analysis was conducted. Airborne images from 2011, 2013, 2015, 2017, 2019, 2021, and 2023 were used in this study. According to the metadata, all images were orthorectified images acquired in April or May and had a spatial resolution of 0.5 m. Image processing was conducted using the geographic information system software QGIS version 3.26.2. The boundaries of the water bodies were manually interpreted and digitized from each temporal dataset, and changes in the surface area of the digitized polygons were then calculated.~~For the airborne image analysis, the boundaries of the water bodies were manually interpreted and digitized from each temporal dataset, and changes in the surface area of the digitized polygons were subsequently calculated.~~

In addition, vegetation surveys and tree-ring analyses were conducted to examine whether hydrarch succession is occurring within the wetland. Drone imagery was acquired on April 5, 2025, using a DJI Mavic 3 platform, and a current vegetation map was produced using QGIS version 3.34.11. The resulting vegetation map was used to determine whether a vegetation gradient exists from aquatic plants to terrestrial herbaceous species, shrubs, and trees.~~This analysis was used to~~

~~determine whether a vegetation gradient exists from aquatic plants to terrestrial herbaceous species, shrubs, and trees.~~ Separately,~~Furthermore~~, on February 19, 2025, increment cores were collected from *Cryptomeria* spp. stands adjacent to the water bodies using an increment borer. Tree-ring analysis of the cores was conducted to evaluate whether a spatial age gradient is present, with tree age increasing with distance from the water bodies (Fig. 1d).

Soil analysis was conducted to assess peat development and organic matter accumulation in the wetland and to provide field-based parameters for ecohydrological modeling. In July 2025, a total of 19 soil cores were collected using a soil core sampler (Fig. 1c). The cores were sectioned at 10 cm depth intervals, resulting in 120 soil samples. The maximum sampling depths of the individual cores ranged from 20 to 80 cm. The collected soil samples were oven-dried at 105 °C for at least 48 h, ground, and sieved through a 2 mm mesh prior to analysis.

Soil organic matter content was determined using the loss-on-ignition method. Dried soil samples were placed in pre-weighed crucibles and weighed before ignition. The samples were then combusted in a muffle furnace at 550 °C for 4 h, cooled to room temperature under laboratory conditions, and reweighed. Organic matter content was calculated as the percentage of mass loss during ignition relative to the pre-ignition dry mass.

To examine whether organic matter content differed among depth intervals while accounting for the non-independence of samples collected from the same soil core, a linear mixed-effects model was fitted with depth interval as a fixed effect and soil core as a random intercept. The model was fitted using R version 4.5.0 with the lme4 and lmerTest packages, and the significance of the depth effect was assessed using Satterthwaite-approximated denominator degrees of freedom.~~soil depths, one-way analysis of variance (ANOVA) was performed using R version 4.5.0. In addition, the coefficient of variation (CV) of organic matter content was calculated across depth intervals within each core to assess vertical variability in organic matter content.~~ The CV was calculated by dividing the standard deviation of organic matter content within each core by the mean organic matter content of that core and multiplying by 100. CV values were classified into least variable (< 15%), moderately variable (15–35%), and most variable (> 35%) classes according to Wilding (1985).

Additional analyses were conducted to evaluate whether geomorphic, ecological, and hydrological processes were interrelated. First, soil sampling sites were classified according to vegetation type into terrestrial vegetation, *Trapa japonica*, and *Schoenoplectiella triangulata*, and one-way ANOVA was performed to test whether organic matter content differed among vegetation types. The organic matter content of each core was calculated as the mean value across all sampled depth intervals. Second, the distance from each soil core to the pond center was calculated in QGIS, and one-way ANOVA was performed to test whether the distance to the pond center differed among vegetation types. The pond center was defined as the centroid of the digitized pond polygon, and the Euclidean distance between each soil core and the centroid was measured. Next, Pearson correlation analysis was conducted to examine the relationship between mean organic matter content and distance from the pond center. For all ANOVA tests, Tukey's post hoc test was conducted to identify pairwise differences among groups.

2.3. Model development

A dynamic Eco–Geo–Hydro model was developed to simulate the annual water budget and pond-area dynamics of the peat wetland by incorporating hydrarch succession and ecogeomorphological processes. All water-budget fluxes, including precipitation, evapotranspiration, direct runoff and subsurface loss~~infiltration~~, were calculated at a daily time step and then aggregated to annual totals for use in the annual simulation. Using the observed pond area in 2011 as the initial condition, the model simulated changes in water storage, water depth, and pond area in Seoyeongari wetland from 2012 to 2023. Based on geomorphological surveys and remote sensing analyses of the wetland, the initial water depth in 2011 was set to 1.2 m, and the initial water storage was calculated using the observed pond area in 2011 and the depth–storage relationship. Water storage was updated at an annual time step using the water-budget method shown in Eq. (1), and pond water depth and area were allowed to vary accordingly with changes in storage (Fig. 2).

$$W = A_P \times (P_P - ET_P \cancel{- T_P} - Q_P) + A_C \times R - A_R \times (ET_R \cancel{+ T_R}) \quad (1)$$

where W = Change in water storage (m³ day^{-1}), A_P = Pond area (m²), P_P = Precipitation on the pond surface, E_P = Evapotranspiration~~Evaporation~~ from the pond (mm day^{-1}), ~~T_P = Transpiration from aquatic vegetation within the pond (mm day^{-1}),~~ Q_P = Infiltration from the pond (mm day^{-1}), A_C = Contributing area (m²), R = Direct runoff from the contributing area entering the pond (mm day^{-1}), A_R = Riparian area (m²), E_R = Evapotranspiration~~Evaporation~~ from the riparian area (mm day^{-1}), and T_R = ~~Transpiration from terrestrial vegetation in the riparian area~~ (mm day^{-1}). All variables and their corresponding symbols are listed in Supplementary Table A.2.

The contributing area was determined using a Digital Elevation Model (DEM) constructed by interpolating 5 m contour data provided by the National Geographic Information Institute (NGII) to a 1 m resolution using Inverse Distance Weighting in QGIS version 3.34.11. The 2011 pond polygon was designated as an internal outlet, and flow directions were derived from the hydrologically conditioned DEM using the D8 algorithm (O'Callaghan and Mark, 1984). The watershed boundary was then delineated by tracing all upslope cells draining to the pond cells (Jenson and Domingue, 1988).

The width of the riparian area was set to the mean width calculated by dividing the total area of the vegetation band directly adjacent to the pond margin by the perimeter of the pond. In the final model, this mean width was set to 17.5 m, and a buffer of this width was applied around the 2011 pond polygon to calculate the riparian-to-pond area ratio. During the simulation, this ratio was multiplied by the simulated pond area in each year to estimate the dynamic riparian area. Pond area and water storage were calculated using depth–area–storage relationships based on the observed pond area in 2011, the initial water depth, and a shape parameter. The shape parameter was set to 1.3 and was used to represent the nonlinear change in pond area with water depth. A relaxation coefficient of 0.035 was applied to water-depth changes so that the target depth derived from the water-balance calculation was not reached instantaneously.

Daily precipitation data (P) from the Seogwang-ri automatic weather station were obtained from the Korea Meteorological Administration Open Weather Data Portal using Automatic Weather System (AWS) records (33°18'16.4"N, 126°18'21.6"E). The interception loss coefficient for pond-surface precipitation (I_P) was set to 13% of total precipitation (Ciężkowski et al., 2018). ~~Therefore, effective precipitation reaching the pond surface (P_E) was calculated as shown in Eq. (2).~~

$$\cancel{P_E = P - I_P = P \times 0.87} \qquad \cancel{(2)}$$

서식 지정함: 글꼴: (한글) +본문 한글(맑은 고딕), (한글) 한국어

서식 있음: 들여쓰기: 첫 줄: 0 글자

Pond evaporation (E_P) was calculated using the Penman equation (Eq. (~~3~~1); Shuttleworth, 1993). Daily meteorological data, including air temperature, wind speed, and precipitation, were obtained from the Seogwang-ri AWS. Sunshine duration data were obtained from the Seogwipo Automated Synoptic Observing Station of the Korea Meteorological Administration (33°14′46.2″N, 126°33′55.1″E).

$$E_P = \frac{\Delta}{\Delta + \gamma} \cdot \left(\frac{R_{n,water}}{\lambda}\right) + \frac{\gamma}{\Delta + \gamma} \cdot \frac{6.43(1 + 0.536u_2)}{\lambda}(e_s - e_a) \qquad (\cancel{3}\underline{1})$$

where E_P = Open-water evaporation (mm day^{-1}), Δ = Slope of saturation vapor pressure curve (kPa °C^{-1}), $R_{n,water}$ = Net radiation at the water surface (MJ m^{-2} day^{-1}), γ = Psychrometric constant (kPa °C^{-1}), u_2 = Wind speed at 2 m height (m s^{-1}), λ = Latent heat of vaporization of water (MJ kg^{-1}), e_s = Saturation vapor pressure (kPa), and e_a = Actual vapor pressure (kPa).

~~The slope of the saturation vapor pressure curve (Δ) was calculated following Allen et al. (1998), as shown in Eq. (4).~~

$$\cancel{\Delta = \frac{4098 e^{0}(T)}{(237.3 + T)^2}} \qquad \cancel{(4)}$$

~~where T = Mean temperature (°C), and $e^{0}(T)$ = Saturation vapor pressure at mean air temperature T (kPa).~~

~~The psychrometric constant (γ) was calculated following Allen et al. (1998), using a fixed latent heat of vaporization of λ = 2.45 (MJ kg^{-1}), as shown in Eq. (5).~~

$$\cancel{\gamma = \frac{C_p P_{atm}}{\varepsilon \lambda}} \qquad \cancel{(5)}$$

~~where C_p = Specific heat at constant pressure (1.013 10^{-3} (MJ kg^{-1} °C^{-1})), P_{atm} = Atmospheric pressure (kPa), ε = ratio molecular weight of water vapor/dry air (0.622).~~

~~Atmospheric pressure (P_{atm}) was estimated as a function of elevation using Eq. (6), following Allen et al. (1998).~~

$$\cancel{P_{atm} = 101.3 \left(\frac{293 - 0.0065 Z}{293}\right)^{5.256}} \qquad \cancel{(6)}$$

~~where Z = Elevation (m).~~

~~Net radiation at the water surface ($R_{n,water}$) was calculated as the difference between net shortwave radiation at the water surface ($R_{ns,water}$) and net longwave radiation (R_{nl}) (Eq. (7); Shuttleworth, 1993).~~

$$\cancel{R_{n,water} = R_{ns,water} - R_{nl}} \qquad \cancel{(7)}$$

~~where $R_{ns,water}$ = Net short-wave radiation at the water surface (MJ m^{-2} day^{-1}), and R_{nl} = Net long-wave radiation (MJ m^{-2} day^{-1}).~~

~~Net shortwave radiation at the water surface ($R_{ns,water}$) was calculated as a function of incoming solar radiation at the surface (R_s) and water-surface albedo (α_{water}), which was set to 0.08 (Eq. (8); Shuttleworth, 1993).~~

$$R_{ns,water} = (1 - \alpha_{water})R_S \qquad (8)$$

~~where α_{water} = Albedo of water surface, R_s = Incoming solar radiation at the surface (MJ m^{-2} day^{-1}).~~

~~Vapor pressure was estimated from daily minimum temperature assuming nighttime saturation ($e_a = e_o(T_{min})$) following Allen et al. (1998), and solar radiation (R_s) was derived from sunshine duration using the Angström equation (Eq. (9)). The Angström coefficients were set to $a_s = 0.25,\ b_s = 0.50$, and cloudiness was represented by the sunshine fraction (n/N).~~

$$R_s = \left(a_s + b_s \frac{n}{N}\right) R_a \qquad (9)$$

~~where R_a = Extraterrestrial radiation (MJ m^{-2} day^{-1}) a_s, b_s = Empirical Angström coefficients, n = Actual sunshine duration (h), and N = Maximum possible sunshine duration (h).~~

~~Extraterrestrial radiation (R_a) for each day of the year and latitude was estimated from the solar constant, inverse relative Earth–Sun distance, solar declination, latitude, and sunset hour angle, as shown in Eq. (10).~~

$$R_a = \frac{24 \times 60}{\pi} G_{sc} d_r (\omega_s \sin\varphi \sin\delta + \cos\varphi \cos\delta \sin\omega_s) \qquad (10)$$

~~where G_{sc} = Solar constant (0.0820 MJ m^{-2} min^{-1}), d_r = Inverse relative distance Earth–Sun, φ = Latitude (radians), δ = Solar declination (radians), and ω_s = Sunset hour angle (radians).~~

~~Net longwave radiation (R_{nl}) is governed by the Stefan–Boltzmann law, which states that longwave radiation emission is proportional to the fourth power of absolute temperature. However, the actual net energy flux is reduced due to atmospheric absorption by water vapor, clouds, and carbon dioxide. Therefore, the Stefan–Boltzmann formulation is empirically corrected to account for humidity and cloudiness, as implemented in Eq. (11) of Allen et al., (1998).~~

$$R_{nl} = \sigma\left(\frac{(T_{max} + 273.16)^4 + (T_{min} + 273.16)^4}{2}\right)(0.34 - 0.14\sqrt{e_a})\left(1.35\frac{R_s}{R_{so}} - 0.35\right) \quad (11)$$

~~where σ = Stefan–Boltzmann constant (4.903×10^{-9} MJ K^{-4} m^{-2} day^{-1}), T_{max}, T_{min}, T = Daily maximum, minimum, and mean air temperature (°C).~~

~~Clear-sky solar radiation (R_{so}) was used to correct for cloudiness. Eq. (12) is applicable where the station elevation is less than 6,000 m, air turbidity is low, and the average sun angle above the horizon is approximately 50° (Allen et al., 1998); all of which are satisfied at the study site.~~

$$R_{so} = (0.75 + 2 \times 10^{-5} Z) R_a \quad (12)$$

~~where R_{so} = Clear-sky solar radiation (MJ m^{-2} day^{-1}), and Z = Elevation (m).~~

~~Saturation vapor pressure ($e_o(T)$) was calculated following Allen et al. (1998), as shown in Eq. (13).~~

$$e_o(T) = 0.6108\, exp\left(\frac{17.27T}{T + 237.3}\right) \quad (13)$$

~~where $e_o(T)$ = Saturation vapor pressure at temperature T (kPa).~~

~~Due to the non-linearity of the saturation vapor pressure-temperature relationship, the mean saturation vapor pressure for a day was computed as the mean of the saturation vapor pressure at daily maximum and minimum air temperatures, rather than the saturation vapor pressure at mean air temperature (Eq. (14)), as the latter would result in underestimation (Allen et al., 1998).~~

$$e_s = \frac{e_o(T_{max}) + e_o(T_{min})}{2} \quad (14)$$

~~According to Allen et al. (1998), humidity data can also be estimated from daily minimum air temperature, as expressed in Eq. (15).~~

$$e_a = e_o(T_{min}) \quad (15)$$

Because the presence of aquatic vegetation can alter open-water evaporation, the effective pond-surface evaporative loss was set to 80% of pond evaporation (Mohamed and Bastiaanssen, 2012). Furthermore, following Lupon et al. (2018), who demonstrated that riparian evapotranspiration constitutes a significant component of catchment water budgets, riparian ET was explicitly incorporated into the model. Riparian evapotranspiration was calculated using the FAO-56 Penman–Monteith method with the dual crop coefficient approach (Allen et al., 1998). Riparian evapotranspiration was parameterized by applying different basal crop coefficients and soil evaporation coefficients for bare ground, grassland, and forest. For each land-cover type i, riparian evapotranspiration was calculated by multiplying reference evapotranspiration by the sum of the basal crop coefficient and the soil evaporation coefficient (Eq. (16)).

$$E_{R,i} + T_{R,i} = \left(K_{cb,i} + K_{e,i}\right)(E_O + T_O) \tag{16}$$

where $E_{R,i} + T_{R,i}$ = Riparian evapotranspiration for land-cover type i (mm day^{-1}), $K_{cb,i}$ = Basal crop coefficient for land-cover type i, $K_{e,i}$= Soil evaporation coefficient for land-cover type i, and $E_O + T_O$ = Reference evapotranspiration (mm day^{-1}).

Forest and grassland were assumed to be fully covered by vegetation; therefore, their soil evaporation coefficients were set to zero. Bare ground was assumed to have no basal transpiration, and its evaporative component was represented by the soil evaporation coefficient. The coefficients were set to $K_{cb,forest}$ = 0.95, $K_{e,forest}$ = 0.00, $K_{cb,grass}$ = 0.90, $K_{e,grass}$ = 0.00, $K_{cb,bare}$ = 0.00, and $K_{e,bare}$ = 0.30.

Reference evapotranspiration was calculated using the FAO Penman–Monteith equation (Eq. (17)) following Allen et al. (1998). For daily calculations, soil heat flux (G) was assumed to be zero.

$$ET_O \cancel{+ T_O} = \frac{0.408\Delta\left(R_{n,land} - G\right) + \gamma \frac{900}{T + 273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \tag{17}$$

where $E_O + T_O$ = Reference evapotranspiration (mm day^{-1}), Δ = Slope of the saturation vapor pressure curve (kPa °C^{-1}), $R_{n,land}$ = Net radiation at the land surface (MJ m^{-2} day^{-1}), G = Soil heat flux density (MJ m^{-2} day^{-1}), γ = Psychrometric constant (kPa °C^{-1}), T = Mean air temperature (°C), u_2 = Wind speed at 2 m height (m s^{-1}), e_s = Saturation vapor pressure (kPa), and e_a = Actual vapor pressure (kPa).

~~Net radiation at the land surface ($R_{n,land}$) was calculated as the difference between net shortwave and net longwave radiation, as expressed in Eq. (18), following Allen et al. (1998).~~

$$\cancel{R_{n,land} = R_{ns,land} - R_{nl}} \qquad \cancel{(18)}$$

~~where $R_{ns,land}$ = Net short-wave radiation at the land surface (MJ m^{-2} day^{-1}), and R_{nl} = Net long-wave radiation (MJ m^{-2} day^{-1}).~~

~~Net shortwave radiation at the land surface ($R_{ns,land}$) was calculated using Eq. (19). The albedo of land cover was set to 0.23, which corresponds to the value for a hypothetical reference crop with an assumed crop height of 0.12 m and a fixed surface resistance of 70 s m^{-1} (Allen et al., 1998).~~

$$\cancel{R_{ns,land} = (1 - \alpha_{land})R_s} \qquad \cancel{(19)}$$

Given that the pond is situated at the base of the Seoyeongari oreum slope within a crater depression, direct runoff from the contributing watershed was included as an input component of the water budget in this study. Direct runoff (R) entering the pond from the contributing area was calculated using the NRCS-CN method (Eq. (20); Michigan Department of Environmental Quality, 2006). This method estimates direct runoff based on the runoff curve number (CN), which reflects soil characteristics, infiltration capacity, and land use/land cover conditions. Seoyeongari oreum belongs to the Jungmun soil series, which corresponds to hydrologic soil group B. Therefore, a CN value of 68, representative of forested areas within hydrologic soil group B in Jeju Island, was applied (Lee et al., 2019). The initial abstraction (I_a), representing interception, surface storage, and infiltration prior to runoff, was assumed to be $0.2S$ following the standard NRCS-CN formulation. When precipitation (P) was less than or equal to the initial abstraction (I_a), direct runoff was set to zero. The potential maximum retention (S) was defined as a function of CN, as shown in Eq. (21).

$$R = \frac{(P - I_a)^2}{(P - I_a + S)} \qquad (20)$$

$$S = \frac{25400}{CN} - 254 \qquad (21)$$

where R = Direct runoff entering the pond from the contributing area (mm day^{-1}), P = Daily precipitation (mm day^{-1}), I_a = Initial Abstraction (mm day^{-1}), S = Potential maximum retention (mm day^{-1}), and CN = Curve Number.

The peat layer of a peat wetland can be divided into the acrotelm, which has relatively high permeability and contains minimally decomposed plant tissues, and the catotelm, which has low permeability and is composed of highly decomposed organic matter. Couwenberg and Joosten (2005) assumed no percolation through the catotelm layer in hydrological models of raised bogs. Similarly, Baird et al. (2008), based on hydrological investigations of the Cor Fochno raised bog in western Wales, reported that outflow through the catotelm accounted for less than 1% of net rainfall due to low hydraulic conductivity at the wetland margins. Because geomorphological surveys of Seoyeongari wetland also revealed a catotelm layer exceeding 1 m in thickness, pond percolation (Q_P) in this model was set to 1% of net precipitation (P_P), as shown in Eq. (22).

$$Q_P = 0.01P_P \tag{22}$$

2.4. Model scenarios

In this study, four scenario models were compared: the Baseline, Hydrosere Only, Eco-Geo Only, and Integrated models. The Baseline model excluded both peat accumulation and hydrosere succession processes. In this model, no pond-bottom rise occurred due to peat accumulation, and newly exposed areas formed through terrestrialization were maintained as bare ground. The Hydrosere Only model incorporated only the hydrosere succession process. In this scenario, newly exposed areas formed by terrestrialization were initially treated as bare ground, converted to grassland after one year, and developed into forest cover after five years. This transition period was set based on the age–distance relationship observed in the dendrochronological analysis conducted in this study. The Eco-Geo Only model included only the ecogeomorphological process, represented by pond-bottom rise due to peat accumulation. In this scenario, raw pond-bottom rise was accumulated at a constant rate of 3 mm yr^{-1} to represent peat accumulation (Temmink et al., 2022). The Integrated model incorporated both peat accumulation and hydrosere succession processes, with the same raw pond-bottom rise rate of 3 mm yr^{-1} applied. In the Eco-Geo Only and Integrated models, the accumulated pond-bottom rise was applied gradually using a bottom-relaxation coefficient of 0.08. To account for peat porosity, a porosity factor of 0.70 was applied when updating the effective storage–depth relationship.

3. Results

3.1. Field experiment

Airborne image analysis showed a decreasing trend in the pond area of Seoyeongari wetland over time (Fig. 3a). The pond area declined by 16.1%, from 2242.97 m^2 in 2011 to 1882.70 m^2 in 2023 (Fig. 3b; Supplementary Table A.3). In the pond, *Trapa* spp., *Schoenoplectiella* spp., and terrestrial herbs such as *Arthraxon hispidus* were identified. Around the outer margin of the pond, a *Cryptomeria* spp. stand, evergreen broad-leaved forest communities including *Quercus acuta* Thunb. and *Neolitsea aciculata* (Blume) Koidz., and deciduous broad-leaved communities such as *Lindera erythrocarpa* Makino were distributed. A vegetation gradient was observed from aquatic plants to terrestrial herbaceous plants, shrubs, and trees, radiating outward from the pond (Fig. 4).

The observed age–distance relationship showed a consistent increase in tree age with increasing distance from the pond margin across sampling points (Fig. 5). Trees located closer to the pond were relatively younger compared to those located farther away. In contrast, the age of the reference point (RP), located relatively far from the pond, was determined to be 55 years (Fig. 1d).

Organic matter content across depths ranged approximately from 32% to 75% among all cores (Fig. 6). A one-way ANOVA of organic matter content across depths showed no statistically significant differences ($p > 0.05$). The coefficient of variation (CV) of soil depth within each core was classified as least variable (< 15%) in 16 cores, while the remaining three cores were classified as moderately variable (15–35%). In contrast, kriging interpolation of mean topsoil organic matter content at 0–30 cm depth using ArcGIS Pro revealed clear lateral spatial heterogeneity within the wetland (Fig. 7).

A one-way ANOVA revealed significant differences in organic matter content and distance from the pond centroid among vegetation types ($p < 0.05$) (Figs. 8a, b). Tukey's post hoc test indicated that all three vegetation types differed from each other in both organic matter content and distance from the pond centroid. Organic matter content was highest in *Trapa* spp., followed by *Schoenoplectiella* spp., and lowest in terrestrial herbs, whereas the distance from the pond centroid increased from *Trapa* spp. to *Schoenoplectiella* spp. and was greatest in terrestrial herbs. In addition, correlation analysis between distance to the centroid and mean organic matter content of cores showed a significant negative relationship ($r = -0.70$, $p < 0.01$) (Fig. 8c).

3.2. Simulation of water budget and pond area according to models

The mean annual air temperature ranged from approximately 14.5 to 16.2 °C, showing a gradual increasing trend over the study period (Fig. 9a). The annual rainfall during the study period showed substantial interannual variability, ranging from approximately 1,100 to 2,100 mm yr^{-1} (Fig. 9b). Pond evapotranspiration varied between approximately 1,000 and 1,150 mm yr^{-1}, with relatively small year-to-year fluctuations compared to rainfall (Fig. 9c). The forest ET ranged from approximately 860 to 960 mm yr^{-1} over the observation period (Fig. 9d). The grass ET ranged from approximately 800 to 900 mm yr^{-1} (Fig. 9e), while the bare soil ET ranged from approximately 265 to 300 mm yr^{-1} (Fig. 9f).

All model scenarios reproduced the overall declining trend in pond area during the study period (Fig. 10; Supplementary Table A.3). Differences among model scenarios were relatively small during the early period but increased toward the later years of the simulation. The normalized root mean square error (nRMSE) values were 1.64% for the Baseline model, 1.44% for the Eco-Geo Only model, 1.22% for the Hydrosere Only model, and 1.11% for the Integrated model (Supplementary Table A.4). Among the four model scenarios, the Integrated model, which incorporated both processes, showed the best performance.

4. Discussion

Airborne image analysis revealed a decrease in pond area over time, indicating that terrestrialization has been occurring in the wetland. As the distance from the pond increased, a vegetation gradient was observed from aquatic plants to terrestrial herbaceous species, shrubs, and trees, which is consistent with the typical pattern of hydrarch succession reported in previous studies (Clements, 1916; Cowles, 1901). Tree-ring analysis further indicated that *Cryptomeria* spp. stands have encroached toward the pond, suggesting that the *Cryptomeria* spp. stand established during the afforestation projects of the 1970s has expanded toward the wetland, consistent with the reference point (RP) (Kim et al., 2025). Taken together, these findings indicate that terrestrialization and hydrarch succession are ongoing in Seoyeongari wetland. These changes should therefore be interpreted not merely as changes in vegetation or hydrology, but as the outcome of interactions among ecological, geomorphic, and hydrological processes.

Soil analysis showed no statistically significant differences in organic matter content among soil depths. The high organic matter content and low vertical variability suggest that organic matter has accumulated relatively consistently within the wetland, and continued peat accumulation may contribute to terrestrialization and the development of hydrarch succession. A significant negative correlation was also observed between distance from the pond centroid and the mean organic matter content of each core. This pattern is consistent with previous findings indicating that disturbances such as woody plant invasion, hydrological fluctuations, and other environmental changes can affect peat accumulation processes (Drollinger et al., 2020). In particular, previous studies have suggested that increased transpiration associated with woody plant encroachment can enhance water loss and accelerate terrestrialization through positive feedback mechanisms (Budny and Benscoter, 2016; Price et al., 2003; Saintilan and Rogers, 2015). Furthermore, the significant differences in distance from the pond centroid among vegetation types are consistent with previous research showing that hydrological stability plays an important role in determining wetland plant community composition and biodiversity (Bridgham and Richardson, 1993; Zoltai and Vitt, 1995). Overall, ecological, geomorphic, and hydrological processes are closely interrelated, and each process may reinforce or weaken the others. Therefore, these processes need to be considered in an integrated manner to more accurately predict patterns of terrestrialization.

Among the four models, the Integrated model showed the highest predictive performance for pond area. The superior performance of this model can be attributed to its explicit representation of the combined effects of hydrarch succession and peat accumulation on terrestrialization. Peat accumulation progressively elevated the pond bottom, thereby reducing water storage capacity and contributing to pond area decline. This mechanism is consistent with the autogenic ecogeomorphological feedbacks conceptualized in models of raised-bog development (Clymo, 1984; Ingram, 1982; Foster and Wright, 1990). Meanwhile, woody plant encroachment increased water loss through enhanced evapotranspiration, further contributing to pond area decline (van der Valk, 1981; Budny and Benscoter, 2016; Saintilan and Rogers, 2015). Previous studies have suggested that vegetation dynamics can modify wetland water budgets and accelerate terrestrialization (Budny and Benscoter, 2016; Price et al., 2003; Saintilan and Rogers, 2015), but most existing water-budget models have not explicitly incorporated these processes (Chen et al., 2021; Mokhesuoe et al., 2023; Wallace et al., 2024). Therefore, the improved accuracy of the Integrated model suggests that explicitly coupling peat accumulation and hydrarch succession within a water-budget framework can provide a more process-explicit representation of the eco-geo-hydrological processes shaping wetland terrestrialization (Budny and Benscoter, 2016; Darlington, 1943; Drollinger et al., 2020; Lindeman, 1941; Price et al., 2003; Saintilan and Rogers, 2015).

This study, however, has several limitations. First, the peat accumulation rate was assumed to be constant, although it may vary depending on climatic conditions, decomposition rates, and vegetation composition (Clymo, 1984). Second, groundwater interactions were not explicitly incorporated into the model, although they may influence hydrological dynamics in certain wetland systems (Baird et al., 2008). Third, uncertainties associated with vegetation parameters may also affect model results (Price et al., 2003). Fourth, all models underestimated the observed rate of wetland area decline after 2020. This underestimation may be related to the high rate of groundwater-level decline in western Jeju Island, where Seoyeongari wetland is located, associated with increasing groundwater use and land-cover change (Jeong et al., 2022).

Future studies should incorporate dynamic peat growth processes, groundwater exchange, and long-term monitoring data to improve model robustness and predictive accuracy. Nevertheless, model performance was improved by integrating these two processes into a water-budget modeling framework that has been applied to individual wetland studies across various regions to predict terrestrialization. Therefore, the model developed in this study may be applicable to other individual wetlands in similar contexts and may help improve predictive performance.

5. Conclusions

This study examined terrestrialization in an individual wetland by applying field surveys and a water-budget model that integrates ecological, geomorphological, and hydrological processes. The field survey results showed the co-occurrence of pond area reduction, a vegetation gradient from aquatic plants to terrestrial vegetation, tree encroachment, and spatial differences in soil organic matter content, suggesting that ongoing terrestrialization in the study wetland is proceeding through interactions among ecological, geomorphological, and hydrological processes.

The model scenario comparison also showed that the Integrated model, which incorporated both peat accumulation and hydrarch succession, most closely matched the observed changes in pond area. Therefore, predicting long-term changes in small wetlands requires consideration not only of external hydrological conditions but also of internal ecogeomorphological and ecohydrological feedbacks within the wetland.

However, because the analysis was based on a single wetland and the model did not explicitly account for long-term water-level observations, groundwater exchange, or temporal variability in peat accumulation rates, further validation using additional field data and other wetland sites is needed. Nevertheless, this study demonstrates that terrestrialization should not be understood simply as a reduction in pond area, but as a coupled process in which ecological, geomorphological, and hydrological processes interact with one another and determine the future state of the wetland. This perspective is significant because it highlights the need to integrate ecological, geomorphological, and hydrological processes in the conservation and long-term prediction of individual wetlands.

CRediT authorship contribution statement

Sungjae Woo: Conceptualization, Investigation, Analysis, Writing – original draft. **Daehyun Kim:** Investigation, Supervision, funding acquisition.

Declaration of competing interest

There is no conflict of interest to be declared.

Data availability

Data will be made available on request.

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